

Short-Circuit Results - 11 kV CYME Feeder

Project: 11 kV radial distribution feeder modeled in CYME/CYMDIST

Study Scope

Short-circuit calculations were used to support the feeder protection-coordination study. The final downstream fault-current values shown below are the values used in the verified sequence-of-operations cases.

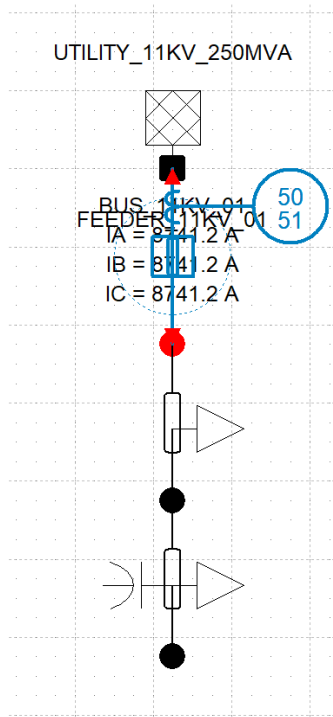
Fault type	Fault current (A)	Verified operating breaker	Trip time (s)
LLL	4,574.7	SEC_11KV_02	0.078
LLG	4,231.4	SEC_11KV_02	0.079
LL	3,961.8	SEC_11KV_02	0.080
LG	3,060.6	SEC_11KV_02	0.085

Upstream Verification

For the upstream test location, the three-phase fault current was approximately **8,741.2 A**. With the final upstream relay time dial of 0.40, breaker **SEC_11KV_01** opened at approximately **0.221 s**.

Quantity	Value
System voltage	11 kV
Downstream full-load current	452.6 A
Downstream minimum fault current (LG)	3,060.6 A
Upstream three-phase fault current	8,741.2 A

Representative CYME Short-Circuit Result



Engineering Note

The fault-current results were subsequently used with the selected CT and relay settings to verify selectivity through TCC and Sequence of Operations analyses. This document records the final portfolio-study results rather than a utility-grade protection study.